

CM12005

‘Calorie-wise’: A PI System for Smart Calorie Tracking

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‘Calorie-wise’: A PI System for Smart Calorie Tracking

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Abstract

Calorie-wise is a Personal Informatics (PI) system designed to help university students track their daily calorie intake efficiently. With the increasing importance of maintaining a balanced diet amidst demanding student life, the system provides users with real-time insights into their eating habits, enabling informed decisions about their nutrition. Students can log the food they eat throughout the day, and the system calculates the calories and macronutrients (like protein, carbs, and fat) automatically.

Calorie-wise integrates manual food logging with calorie and macronutrient calculation, exercise logging, water intake tracking, and an intuitive dashboard displaying daily progress and trends. The system allows users to set personalised calorie goals and track how close they are to meeting those goals. It also provides ways to log exercise activities and see how those activities impact their overall calorie balance. Users can track how much water they drink each day, which is another key part of a healthy lifestyle. Features considered for future development include meal recommendations, perhaps based on a student's budget or dietary preferences, barcode scanning for faster food entry, and more detailed settings customisation.

The development process followed an Agile methodology, specifically the Scrum framework, structured around two sprints, which allowed for iterative improvements based on team feedback and evolving understanding. We focused on getting a working version out quickly in the first sprint, and then added more features in the second sprint. This report outlines the problem domain analysis, why students need help with tracking their food, the software development lifecycle (SDLC) approach followed, the system's requirements and design, the verification strategy through testing, and a critical reflection on both the product and the process, highlighting key learnings and areas for future enhancement.

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1. Introduction

Problem Area: Personal Informatics

Personal Informatics (PI) refers to systems designed to help individuals collect personal data and reflect on it to gain self-understanding and promote positive habits (Li, Dey and Forlizzi, 2010). PI tools are widely adopted in health and wellness, from fitness trackers to sleep apps. However, their effectiveness hinges on aligning with user routines and motivations. Often, generic tools fail to address the specific struggles faced by particular user groups, such as university students.

University students frequently struggle with maintaining a balanced diet due to irregular schedules, budget constraints, and limited cooking facilities (Pelletier and Laska, 2012). Poor dietary habits can negatively impact physical health, cognitive performance, and increase stress and anxiety (Raza, 2021). While many calorie-tracking apps exist, they are often not tailored to student needs like quick logging, budget considerations, and relevant nutritional guidance combined in one accessible platform.

The Gap and Our Solution

Successful PI systems must balance usability, personalisation, and actionable insights (Epstein et al., 2015). Many existing applications overwhelm users with complex interfaces or premium feature subscriptions, or fail to provide contextually relevant recommendations, particularly for students. Our software system, Calorie-wise, addresses these gaps by offering a student-centric approach to calorie and basic health tracking.

The main idea behind Calorie-wise is to provide university students with an intuitive, web-based tool that:

- Simplifies food logging with calorie and macronutrient tracking.
- Allows tracking of water intake and exercise expenditure.
- Presents progress visually to aid understanding and motivation.

By focusing on student needs identified through research, our system aims to make health tracking more accessible and sustainable, ultimately promoting healthier habits within this demographic.

Background Research

To better understand student dietary behaviours and needs, we surveyed 30 University of Bath students (see Appendix 3 for details). As PI systems aim to help users track personally relevant data for reflection and informed behavioural change (Li, Dey and Forlizzi, 2010),

our goal was to discover current behaviours and identify where support was most needed from a student perspective. We aimed to analyse current dietary habits while understanding willingness to engage with a PI tool for reflecting on health habits.

Findings revealed that while 83.3% of students have fitness goals, only 6.7% actively use a PI system for calorie intake, and 36.7% have used tools inconsistently. This discrepancy may stem from the perceived complexity of modern PI tools and busy academic schedules making tracking seem overwhelming (Li et al., 2012). Furthermore, a lack of personalisation could be a barrier; effective PI systems should align with user routines and needs (Epstein et al., 2015). For instance, 46.7% of students have food budgets, an external factor the app must consider.

Despite fitness goals, 80% regularly skip meals, and 50% rated their diet/lifestyle satisfaction $\leq 5/10$. These findings parallel students facing time/budget constraints impacting healthy eating (Pelletier and Laska, 2012), supported by 43.3% getting takeout 2-3 times weekly. Open responses regarding diet challenges included 'Finding time to cook', campus food being 'Overpriced', and needing 'Proper meals'. Therefore, Calorie-Wise must consider these challenges and offer personalisation to be effective long-term (Epstein et al. 2015).

Including tracking streaks was a popular feature request, highlighting interest in behavioural reinforcement common in PI design for self-reflection and gaining self-knowledge (Rapp and Cena, 2014). Streak tracking is a form of self-monitoring which can lead to desired behavioural changes, though effects can be limited by maintaining motivation and accuracy (Rapp and Cena, 2014). To address this, streaks should reinforce consistency paired with positive feedback, promoting long-term engagement rather than becoming a chore, which would defeat the app's purpose.

2. Agile Software Process Planning and Management

Our team adopted the Agile methodology, specifically the Scrum framework, to manage the development of Calorie-Wise. This approach allowed for flexibility, iterative development, and continuous feedback integration.

Scrum Planning and Tracking

The project commenced with an initial planning session led by the Product Owner (Tahmid). We defined the project vision, identified key stakeholders (university students), and established an initial Product Backlog containing high-level features derived from the project goals and initial research, such as user authentication, calorie tracking, water logging, exercise logging, and data visualisation.

We utilised Jira for managing the Product Backlog and planning our sprints. Each backlog item (User Story, Task, Bug) was assigned a priority, estimated (where applicable), and allocated to sprints based on team capacity and dependency considerations. Items selected for a sprint formed the Sprint Backlog. These items were further broken down into smaller, manageable subtasks (e.g., design UI element, implement backend logic, write test case).

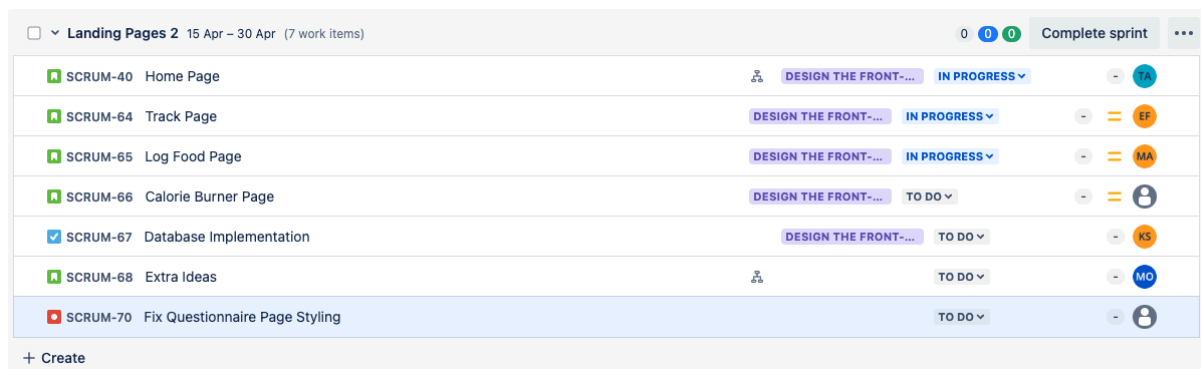


Figure 2.1.1: Jira Sprint Backlog

Figure 2.1.1 shows an example of our Sprint Backlog in Jira during Sprint 2. User Stories (green icons, e.g., SCRUM-40 Home Page) represent new features. Tasks (blue icons, e.g., SCRUM-67 Database Implementation) denote specific technical work or research. Bugs (red icons, e.g., SCRUM-70 Fix Questionnaire Page Styling) represent issues identified in previous iterations requiring resolution. The Product Owner continuously reviewed and re-prioritised the Product Backlog based on ongoing research, user feedback, and technical feasibility assessments, ensuring the development effort remained aligned with the most valuable objectives.

Scrum Meetings

Regular Scrum ceremonies facilitated communication, coordination, and adaptation. The Scrum Master (Abdollah) coordinated the team, which was organised into five pairs (subgroups) focusing on specific application areas.

- **Daily Scrums:** Held briefly each day via Discord. Each subgroup reported progress made since the last meeting, planned work for the current day, and identified any impediments. This ensured transparency and allowed for rapid problem-solving.
- **Sprint Planning:** Conducted at the beginning of each sprint. The team collaboratively selected items from the Product Backlog to form the Sprint Backlog, clarified requirements, and defined the Sprint Goal. Tasks were assigned to subgroups.
- **Sprint Reviews:** Held at the end of each sprint. The team demonstrated the completed work (increment) to the Product Owner and potentially other stakeholders (like tutors during lab sessions, as per the spec). Feedback was gathered and used to inform the Product Backlog for future sprints.
- **Sprint Retrospectives:** Conducted after the Sprint Review. The team reflected on the sprint process itself – what worked well, what challenges were faced, and what improvements could be made to enhance collaboration, efficiency, and quality in the next sprint.

Features Chosen

We completed two main development sprints:

- **Sprint 1 (1 week):** Focused on establishing the foundational architecture and core UI elements for a Minimum Viable Product (MVP). Key deliverables included user registration (FR1), login (FR2), the initial questionnaire flow (FR4), and the basic layout/styling for the Login, Register, Questionnaire, and Home pages (NFR5). GitHub branches were used for parallel development by subgroups, merged at the end of the sprint.
- **Sprint 2 (3 weeks):** Focused on implementing core tracking functionality and data display. Key deliverables included manual food logging with calorie/macro calculation (FR5, FR6, FR8), water logging (FR9, FR10), manual exercise logging (FR12), displaying daily totals and basic visualisations on the dashboard (FR14), and implementing goal setting/modification (FR5).

Conclusion

In conclusion, Scrum provided valuable structure for the Calorie-wise project, aiding coordination and task management. However, we faced scheduling conflicts and communication challenges, impacting team spirit and task clarity. This revealed the necessity for more proactive team-building activities and potentially utilising more sophisticated communication tools beyond just Discord. Future improvements include stricter meeting schedules, clearer task definitions, and team-building activities to enhance our Agile implementation and teamwork. Despite challenges, we'd reuse Agile but with these refinements.

3. Software Requirements Specification

3.1: Establishing Detailed System Requirements

Developing a successful Personal Informatics (PI) system requires a clear understanding of user needs and iterative refinement based on feedback from the Product Owner. Our calorie and health tracking website, "Calorie-Wise," initially focused on basic calorie tracking. Through user research targeted at university students (our primary user base), this concept evolved into a more comprehensive health tool.

Initial Concept:

We began by considering core functionality:

- A simple interface for logging food items and their calories.
- A display showing the daily total calories consumed.
- A static daily calorie goal for user reference.

While functional, we recognised that achieving user adoption demanded deeper personalisation and tracking of additional health metrics relevant to students.

User Research and Feedback:

To guide refinement, we conducted surveys and interviews with university students (details in Appendix 3). Key insights revealed the need for:

1. **Goal Flexibility:** Students desired personalised goals (e.g., weight loss, maintenance, muscle gain) and the ability to change these goals over time.
2. **Holistic Health Tracking:** Interest extended beyond calories to include water intake and exercise logging.
3. **Motivational Feedback:** Users preferred visual progress summaries (graphs, charts) over static numbers to stay engaged and motivated.

Refined System Requirements:

Based on this feedback, we expanded the system requirements:

- **Dynamic Goal Setting:**
 - An initial questionnaire captures user metrics (age, weight, height, activity level, goal) to automatically calculate a suggested daily calorie goal (e.g., using the Mifflin-St Jeor formula).
 - Users can manually set or adjust their calorie, water, and potentially exercise goals via the settings page.
- **Multi-Metric Tracking:**
 - **Food/Calories:** Manual logging of food items with associated calories and macronutrients.
 - **Water Intake:** Simple logging (e.g., per cup/bottle) with a visual progress indicator against a daily goal.
 - **Exercise:** Manual logging of activities and estimated calories burned, adjusting the net daily calorie balance.

- **Data Visualisation:**
 - Daily summaries presented visually (e.g., pie charts for macros, progress bars for goals).
 - Historical trends (e.g., weekly/monthly views of calorie intake, water consumption) are displayed using graphs.
- **User Experience and Engagement:**
 - Intuitive navigation and clear presentation of information.
 - Features designed to encourage consistent use, such as tracking streaks.

Validation and Iteration:

Initial prototype testing with students led to further refinements incorporated into the requirements, such as:

- Implementing secure user accounts for personalised access across devices.
- Adding a streak counter to gamify progress and encourage daily interaction.

3.2 System Domain and Target Users

Our PI system operates within the health and wellness technology domain, encompassing concepts like the Quantified Self and wearable computing (Swan, 2012), focusing specifically on nutritional awareness and activity tracking for university students. Unlike generic calorie counters, Calorie-Wise is tailored to address the unique context and challenges of this demographic.

Key Domain Characteristics Addressed:

- **Nutritional Tracking:** Provides calorie and macronutrient monitoring adaptable to potentially irregular student eating patterns.
- **Holistic Health Metrics:** Tracks hydration and exercise alongside nutrition, acknowledging their interconnected role in wellbeing.
- **Behavioural Motivation:** Incorporates data visualisation and goal reinforcement features (like streaks) to encourage consistent engagement, addressing common motivation challenges.

Target Audience: University Students

Our primary users are university students (typically 18-25 years old). User research identified key needs and challenges:

- **Challenges Faced:**
 - **Time Constraints & Irregular Schedules:** Academic pressure and social life often lead to skipped meals or reliance on convenience foods.
 - **Budget Constraints:** Limited finances impact food choices and willingness to pay for health apps.
 - **Health Awareness Gaps:** Lack of time or knowledge can lead to a poor understanding of nutritional values.
 - **Privacy Concerns:** Increased awareness and scepticism regarding personal health data collection exist, with studies indicating privacy concerns act as a significant barrier to Gen Z's adoption of health data systems (Al-Khawlani et al., 2023).
- **How Calorie-Wise Addresses These Needs:**
 - **Efficient Tracking & Education:** Provides quick logging features and readily

available nutritional information for logged foods.

- **Free Access:** Core functionalities are offered without subscription fees.
- **Privacy-Centric Design:** User data is secured through individual accounts with passwords, and personal health data is not shared with third parties (NFR3).

3.3 Use Cases

Actor	Student
Goal	Record a meal to track the daily caloric intake
Preconditions	The user is logged in
Basic workflow	1. User selects “Log Food” 2. System displays meals for the day: “Breakfast”, “Lunch”, “Dinner”, “Snacks” 3. Plus button next to each one to indicate adding to which meal they are choosing to log food into 4. The user is given a popup window that allows them to select food 5. Upon selecting food, they can choose the portion size of that food between ounces/pounds/grams/kilograms. 6. System retrieves calories and macronutrients, updating the daily calorie total and confirming food log
Postconditions	Meal is recorded in the user’s daily log, and the calorie progress bar updates

Actor	Student
Goal	Log water intake
Preconditions	The user is logged in
Basic workflow	1. The user navigates to “Water intake” 2. System displays a “+” and “-” button to indicate adding or decreasing cups of water the user drank. 3. User can edit goal 4. System saves preferences and starts tracking
Postconditions	Meal is recorded in the user’s daily log and the calorie progress bar updates

Actor	Student
Goal	Log Exercise
Preconditions	The user is logged in

Basic workflow	1. The user navigates to “Log exercise” 2. System displays a way for user to log the number of calories they have burnt on their own 3. User confirms their entry 4. System adjusts the limit for calories burned and saves it in the system
Postconditions	Meal is recorded in the user’s daily log and the calorie progress bar updates

Table 3.3.1: Goal Use Cases

3.4 Functional and non-functional requirements

	3.4.1 Functional Requirements	Priority
User Management	FR1: The system shall allow users to create an account and login using an email/username and password.	High
	FR2 :The system shall allow users to reset forgotten passwords, or remember passwords when logging in.	Medium
	FR3: The system shall determine the users caloric goals based on the questionnaire.	High
	FR4: The System shall allow the user to change their caloric goals at any time.	Medium
Calorie Tracking	FR5: The system shall allow manual entry of food items with calorie values	High
	FR6: The system shall calculate and display daily calorie totals	High
	FR7: The system shall compare daily intake against user-set goals	High
	FR8: The system shall also track macronutrients along with calories.	Medium
Water Tracking	FR9: The system shall provide quick log button for cups	High
	FR10: The system shall display water intake	High
Exercise Tracking	FR11: The system shall be able to link a tracking device to be able to track calories burned through steps.	Low (not implemented)
	FR12: The system shall be able to take a users input of how many hours or calories they have burned and update the main system	High

Data Visualisation	FR13: The system shall have graphs showing calories consumed average daily, over a month	High
	FR14: The system shall have graphs showing calories burned average daily, over a month	High
	FR15: The system shall allow users to view historical trends in calorie intake and calories burned across multiple days	Medium
	FR16: The system shall display a visual streak counter for consecutive days of system usage to encourage habit formation and reflection	Medium

Table 3.4.1: Functional Requirements

3.4.2 Non-Functional Requirements:		Priority
Data and usability	NFR1: The system shall have a working database that stores user details and additional information about the user.	High
	NFR2: The system shall be able to run completely locally and offline	Low
	NFR3: No personal data shall be shared with third-party advertisers.	High
	NFR4: Web version shall work on Chromium Browsers, Safari, and Edge (latest 2 versions).	Medium
	NFR5: The system shall provide a dashboard for easy access, there should be a consistent layout/styling including access to other page and a settings page	High

Table 3.4.2: Non-functional Requirements

4. Design

This section details the design of the "Calorie-Wise" Personal Informatics system. Our approach was iterative, guided by our agile approach, user research insights (detailed in Section 3.1), and established PI design principles to create a tool tailored for university students. The design focuses on usability, flexibility, and providing motivational feedback.

4.1 Design overview

The system employs an object-oriented design implemented as a web application. Key architectural components include:

1. **User authentication module:** Handles user registration, login, and session management securely.
2. **Core tracking modules:** Separate components for managing calorie/macronutrient, water, and exercise data logging and calculations.
3. **Goal management module:** Allows users to set and adjust personal health goals based on initial questionnaire data or update it manually.
4. **Data persistence layer:** A database stores user profiles, logged data, goals, and streak information (NFR1).
5. **User interface (UI):** A dashboard-centric frontend provides access to logging functions, data visualisations, and settings, designed for consistency and ease of use (NFR5).

The design evolved across sprints, starting with foundational elements like authentication and basic layout in Sprint 1, followed by core tracking logic and data visualisation implementation in Sprint 2.

4.2 UML models

The following UML models describe the low-level design of the Calorie-Wise system.

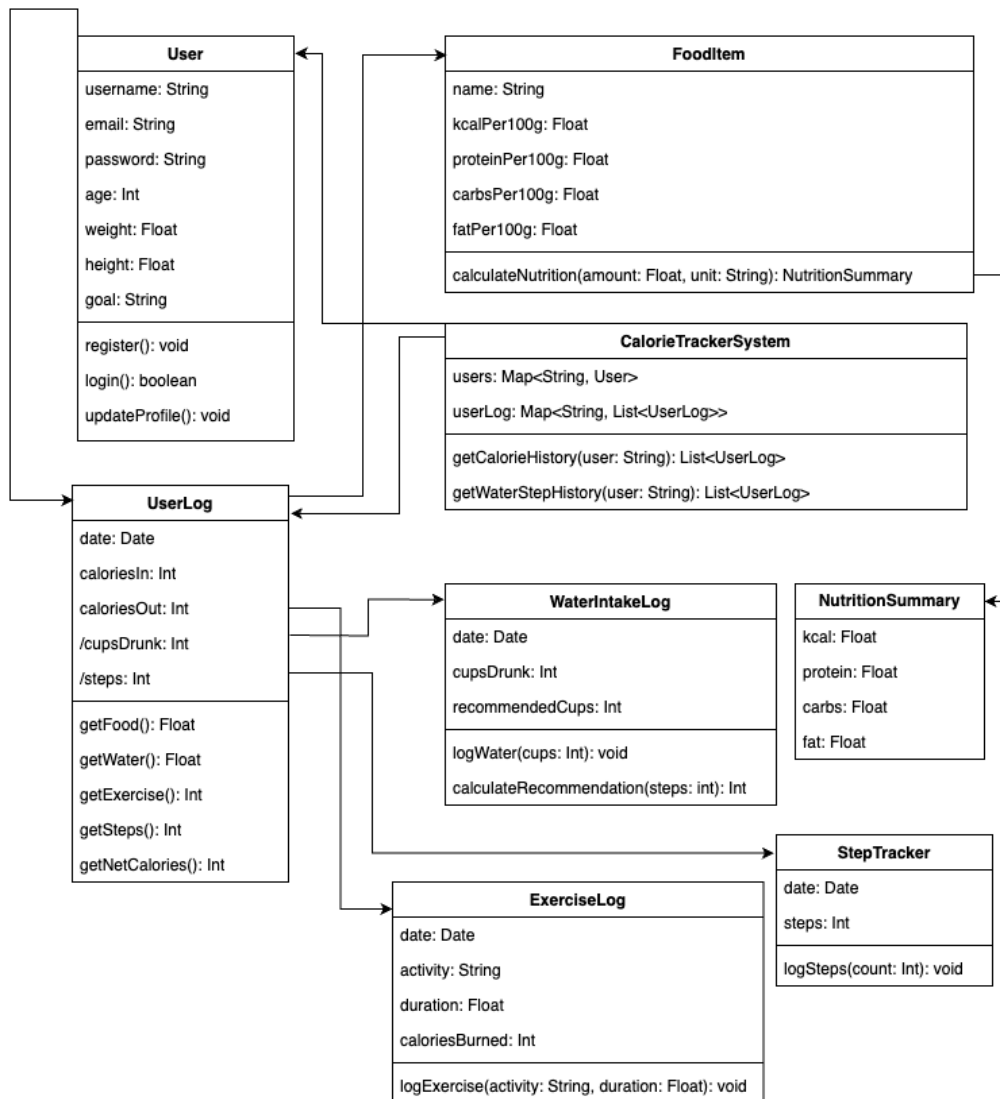


Figure 4.2.1: System class diagram

Figure 4.2.1 illustrates the static structure of the Calorie-Wise system, outlining the main classes, their attributes, methods, and relationships. The **User** class encapsulates profile data and authentication methods (FR1, FR2, FR3, FR4). **FoodItem**, **WaterIntakeLog**, **ExerciseLog**, and **StepTracker** manage specific data points (FR5, FR8, FR9, FR10, FR12). **UserLog** aggregates daily data, while **CalorieTrackerSystem** acts as a central manager. This modular structure supports functional requirements and separates concerns for maintainability. (Ensure explanatory text matches the final diagram precisely.)

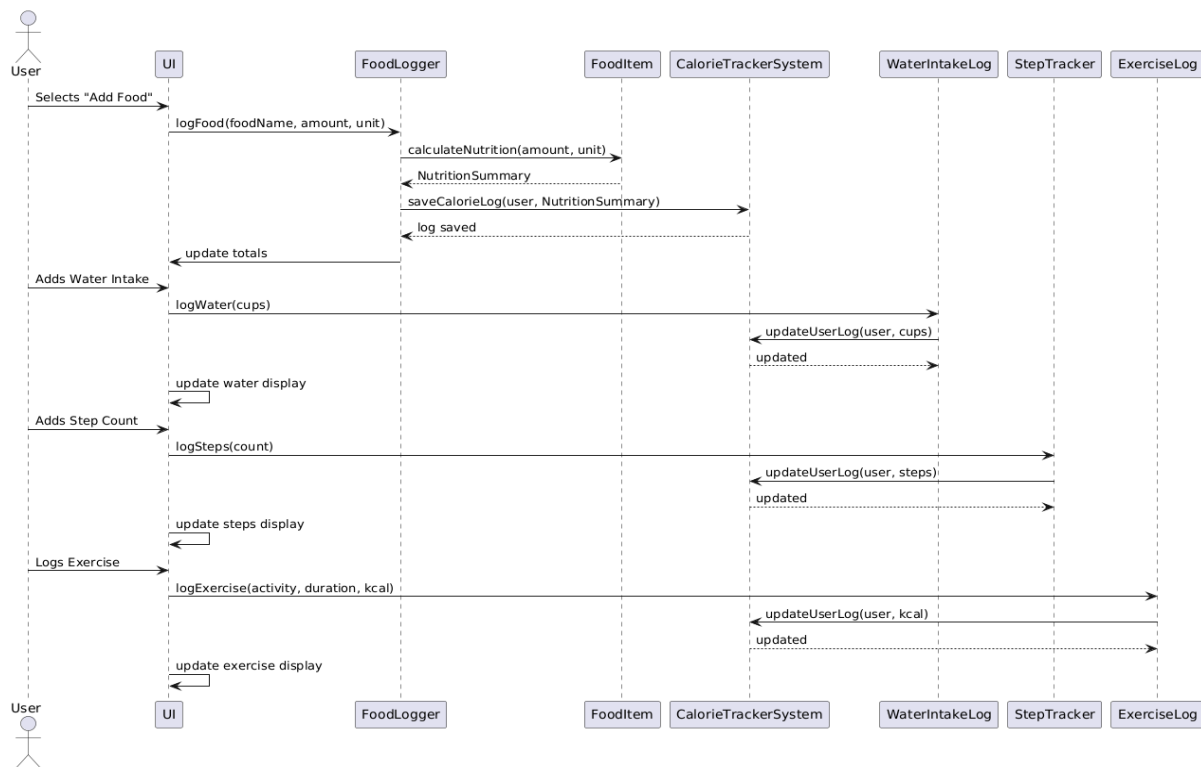


Figure 4.2.2: Sequence diagram

Figure 4.2.2 details the dynamic interactions for several key logging workflows initiated by the User via the UI. It illustrates the common pattern where user actions (logging food, water, steps, or exercise) trigger calls to specific handler objects (FoodLogger, WaterIntakeLog, StepTracker, ExerciseLog). These handlers process the input, potentially interacting with other classes like FoodItem for nutritional calculation. Subsequently, the central CalorieTrackerSystem is invoked to update the appropriate user's daily UserLog instance. Finally, feedback is provided to the UI to update the display. This visualisation clarifies how components collaborate to fulfil core tracking requirements (FR5-FR13) and ensures data is consistently processed and aggregated for user reflection.

4.3 Design justification and PI principles

Our design choices were driven by the goal of creating an effective PI tool specifically for students, grounded in user research and PI literature. The following table summarises the justification for key design aspects and their link to requirements:

Design Area	Key Design Aspects	Justification (PI Principles & Requirements Link)
User Authentication & Personalisation	Secure accounts (email/password login: FR1, FR2; reset). Initial questionnaire for calculated goals (FR3). Manual goal setting/adjustment (FR4, FR5).	Personal accounts essential for tracking <i>personal</i> data over time (NFR1). Questionnaire addresses the user's need for goal flexibility. Manual adjustments enhance Personalisation (key for PI effectiveness - Epstein et al., 2015). Clear UI ensures ease of access (NFR5).

Dashboard-Centric UI & Navigation	Primary UI is a dashboard showing key metrics. Consistent layout/styling (NFR5) across pages. Settings access via icon (FR4, <i>NFR5</i>).	Dashboard provides immediate overview, supporting Self-Reflection (Li, Dey and Forlizzi, 2010). Centralised functions minimise navigation complexity (NFR5). Consistent styling ensures cohesive experience. Initial layout established dashboard concept, prioritising intuitive information access.
Multi-Metric Tracking Modules	Dedicated functions for logging calories/macros (FR5, FR6, FR8), water (FR9, FR10), exercise (FR12). Food logging uses DB (FR7, <i>NFR1</i>), compared to goals (FR7, FR13, FR14). Water uses increments (FR9). Exercise adjusts net calories (FR12).	Addresses user desire for Holistic Health Tracking. Modular design aids development/expansion. Flexibility (manual entry) balanced with efficiency (quick water log). Provides detailed nutritional awareness + broader health factors. Core logic implemented in Sprint 2.
Data Visualisation & Motivation	Daily summaries via charts/progress bars and historical trends via graphs (FR13, FR14, FR15). Streak counter for goal achievement (FR16).	Addresses user preference for Motivational Feedback. Visualisations aid interpretation and Reflection (Li, Dey and Forlizzi, 2010). Progress bars give immediate feedback. Streaks use Gamification to encourage consistency (Rapp and Cena, 2014). Daily visualisation implemented in Sprint 2.
Privacy & Accessibility	Secure data storage, no third-party sharing (<i>NFR3</i>). Cross-browser compatibility (<i>NFR4</i>). Offline access considered (<i>NFR2</i> - low priority).	Addresses student Privacy Concerns. Cross-browser compatibility maximises accessibility.

Table 4.3.1: Design Justification

4.4 Requirement Traceability

This table links key design features/components to the requirements they primarily satisfy:

Design Feature/Component	Functional Requirements (FR IDs)	Non-Functional Requirements (NFR IDs)
User Authentication UI	FR1, FR2	NFR1, NFR5
Questionnaire Module	FR3	NFR1, NFR5
User/Goal Settings Page	FR4	NFR5
Food Logging Interface/Logic	FR5, FR6, FR7, FR8	NFR1, NFR5
Water Logging Interface/Logic	FR9, FR10	NFR1, NFR5
Exercise Logging Interface/Logic	FR11, FR12	NFR1, NFR5
Dashboard UI (Layout/Styling)	FR3, FR5, FR6, FR7, FR8, FR10,	NFR1, NFR5

	FR13, FR14	
Daily Data Visualisation Components	FR13, FR14	NFR1, NFR5
Historical Data Graphs	FR15	NFR1, NFR5
Streak Counter Logic/Display	FR16	NFR1, NFR5
Database Schema/Access Layer	FR1-FR16	NFR1, NFR3, NFR4
Core CSS Framework	FR5, FR6, FR8, FR9, FR10, FR12, FR13, FR14, FR15, FR16	NFR4, NFR5

Table 4.4.1: Requirement Traceability

Our system design was intentionally modular to support flexibility, future extensibility, and alignment with PI principles. Separating data capture from aggregation follows the Single Responsibility Principle and allows for potential future integrations. This structure ensures the system is robust yet adaptable.

5. Software Testing (Verification)

5.1 Testing Plan

Testing will be conducted at the end of each sprint. Usability testing will be carried out via a focus group, which will be a group of around three to five people who will test, evaluate, and provide feedback on our game throughout the stages of development. The members of this focus group will represent the stakeholders for our solution.

System testing will be carried out at the end of each sprint, once a final executable version of the program is produced. Any tests which fail, will be carried over as tasks in subsequent sprint. Refer to Figure 2.1.1, which demonstrates a task being carried over (at the bottom).

5.2 Sprint 1

ID	Description	Input	Expected output	Actual output	Justification
TC-1	Verify successful login with correct credentials	Enter valid username/password, click login	User redirected to Homepage	As expected	To verify core login functionality (FR1, NFR1)
TC-2	Verify successful user registration	Enter valid registration details, click Register	Account created, user redirected (e.g., to login/questionnaire)	As expected	To verify core registration functionality (FR1, FR2, NFR1)
TC-3	Verify questionnaire page loads correctly	Navigate to questionnaire page (e.g., after register)	Questionnaire page elements (questions, inputs) displayed	As expected	To verify questionnaire page rendering (FR3, NFR1)
TC-4	Verify homepage layout renders correctly	Log in successfully	Homepage shows dashboard area	As expected	To verify (FR1, NFR4, NFR5) is met
TC-5	Verify settings button is visible and positioned	View homepage	Settings button displayed	As expected	To verify (FR1, NFR5) is met
TC-6	Verify consistent styling across login, register, questionnaire, and home pages	View login, register, questionnaire, and home pages	Fonts, colours, spacing, layout elements match style Guide consistently across pages	Consistent styling, as expected	To confirm adherence (NFR5) across pages

Table 5.2.1: Testing for Sprint 1

5.3 Sprint 2

ID	Description	Input	Expected output	Actual output	Justification
TC-7	Verify manual food entry calculates and displays calories correctly	Log food item "Apple" with quantity "1 medium", click Save	Daily calorie total increases by expected amount (e.g., 95 kcal), item appears in daily log.	As Expected	To verify core calorie tracking functionality (FR5, FR6).
TC-8	Verify macronutrient data is displayed after logging food	Log food item with known macros (e.g., "Chicken Breast"), view log	Daily macro totals (protein, carbs, fat) update and are displayed correctly.	As Expected	To verify macronutrient tracking (FR8).
TC-9	Verify daily calorie total compares correctly against goal	Log food items exceeding the daily goal	Visual indicator (e.g., progress bar turns red, text indicator) shows goal exceeded.	As Expected	To verify goal comparison logic (FR7).
TC-10	Verify water intake logging increments correctly	Click "+" button for water intake twice	Water intake display shows "2 cups" (or equivalent units).	As Expected	To verify water logging mechanism (FR9, FR10).
TC-11	Verify manual exercise entry adjusts net calorie goal	Log exercise "Running, 30 mins, 300 kcal burned"	Net calorie allowance for the day increases by 300 kcal (or equivalent display adjustment).	Returns original values. It does not update the values stored	To verify manual exercise logging and net calorie calculation (FR12).
TC-12	Verify dashboard visualization updates after logging food	Log a food item, view dashboard	Dashboard chart/graph/summary reflecting calorie/macro intake updates immediately.	No change, placeholder chart still shown	To verify data visualization reflects current data (FR15).
TC-13	Verify dashboard visualization updates after logging water	Log water intake, view dashboard	Dashboard element showing water progress updates immediately.	No visual bar update, only numbers change	To verify data visualization reflects current data (FR10, FR14, NFR5).
TC-14	Verify user can change calorie goal in settings	Navigate to Settings, change calorie goal to new value, click Save	Return to the dashboard, the new calorie goal is displayed and used for comparison.	Returns original values. It does not update the values stored	To verify goal modification functionality (FR4, NFR5).
TC-15	Verify food database lookup	Start typing known food item (e.g., "Banana") in food log search	System suggests relevant food items from the database with calorie/macro info.	As Expected	To verify food database integration (FR5, FR6, FR8, NFR1).

Table 5.3.1: Testing for Sprint 2

6. Reflection and Conclusion

6.1 Critique of the System

6.1.1 Requirements specification Critique

- **Strengths:**
 - Functional requirements (FRs) were systematically identified, categorized (User Management, Tracking, etc.), and prioritised, providing a clear scope (Section 3.4.1).
 - Requirements were derived from initial goals and refined based on user research (surveys/interviews), aligning the system with target user needs (Section 3.1).
 - Non-functional requirements (NFRs) addressed key areas like data storage (NFR1), privacy (NFR3), and compatibility (NFR4) (Section 3.4.2).
- **Areas for Improvement:**
 - **Specificity:** Some FRs could be more specific regarding data validation rules (e.g., for questionnaire inputs, food entry values). Security requirements beyond basic login could be detailed further (e.g., password complexity, session handling).
 - **Completeness:** NFRs could be expanded to explicitly cover aspects like performance targets (this goes beyond NFR5), scalability considerations for the database beyond basic functionality, and detailed usability heuristics. Assumptions made (e.g., user knowledge of nutrition) could be explicitly stated.
 - **Traceability:** While traceability was attempted (Section 4.4), consistently linking all design elements and test cases back to specific requirement IDs needs rigorous checking.

6.1.2 System design Critique

- **Strengths:**
 - The design employs modularity (Authentication, Tracking, Persistence layers - Section 4.1, Figure 4.2.2), promoting separation of concerns and potential extensibility.
 - Object-oriented principles were applied (Class Diagram, Figure 4.2.1), organising data and behaviour logically.
 - The dashboard-centric UI design (Section 4.3.2) aligns with PI principles for providing users with an accessible overview for self-reflection.
- **Areas for Improvement:**
 - **Database Implementation:** The chosen database implementation (.txt file) proved insufficient for handling real-time updates reliably across different parts of the application. This likely contributed to the failures observed during testing, where dashboard visualisations did not update correctly after logging food or water (TC-12, TC-13), and changes to user goals made in settings were not reflected back on the dashboard (TC-14). A more robust database system (e.g., SQL or NoSQL) with proper transaction handling would be required for reliable data consistency.

- **Exercise Calculation Implementation:** The logic for adjusting the net calorie goal based on manually logged exercise (FR12) was not implemented correctly, as evidenced by the failure of test case (TC-11) where the net calorie value did not update as expected. This points to a flaw in the implementation of the calculation or data update mechanism within the Exercise Logging module or UserLog.
- **Data Visualisation Implementation:** While the design included data visualisations (FR13, FR14, FR15, FR16), the implementation in Sprint 2 was incomplete or faulty. Test cases (TC-12) and (TC-13) showed that dashboard charts and progress bars did not dynamically update with new data, instead displaying static placeholder content or only updating numerical values without visual changes. This indicates issues in the frontend JavaScript responsible for fetching updated data and re-rendering the visualisation components.
- **Error Handling:** While we might have addressed this, the design documentation should explicitly detail how errors (e.g., invalid input, database connection issues) are handled to ensure system resilience.
- **Integration Design:** The design for integrating external devices (FR11) was not fully realised or tested, indicating a gap between the requirement (even if low priority) and the implemented design.
- **UI/UX Refinement:** While basic usability was considered (NFR5), advanced UX considerations like intuitive data entry flows and feedback mechanisms for failed actions could be more deeply integrated into the design documentation.

6.1.3 Testing Critique

- **Strengths:**
 - A sprint-based testing approach was adopted, aligning with the Agile methodology (Section 5.1).
 - Test cases were documented with clear inputs, expected outputs, and links to requirements (Tables 5.2.1, 5.3.1).
 - Different levels of testing (sprint/functional, usability, system) were planned (Section 5.1).
- **Areas for Improvement:**
 - **Test Coverage:** It's unclear if the documented test cases provide sufficient coverage for all requirements, including edge cases and error conditions. A requirements traceability matrix mapping tests back to FRs/NFRs would demonstrate coverage more clearly.
 - **Usability and System Testing:** We should have gone through with the planned usability (focus group) and comprehensive system testing.
 - **Test Automation:** The reflection notes the lack of automated testing (Section 6.2.3). Implementing unit tests or UI automation could significantly improve efficiency and regression detection, especially given the coordination challenges mentioned.
 - **Non-Functional Testing:** Explicit test cases for NFRs (e.g., performance under load, compatibility checks on all specified browsers - NFR4, security vulnerability checks - related to NFR3) were not complete.

6.2 Group process reflection

6.2.1 Agile in Practice

Our team successfully adopted Agile principles, primarily Scrum, throughout the project lifecycle. Working in defined sprints with clear goals allowed for iterative development and regular delivery of functional increments. The use of Jira (Figure 2.1.1) was effective for backlog management, task tracking, and visualising sprint progress. Daily Scrums facilitated communication and rapid identification of blockers within subgroups. Sprint Reviews provided opportunities for feedback (primarily internal and from tutors), while Sprint Retrospectives (though perhaps informal) helped us identify process issues, such as the coordination challenges mentioned below. The ability to adapt the backlog based on ongoing understanding (e.g., refining requirements after user research) demonstrated agility in practice. What worked particularly well was the use of Jira for task visibility and the structure provided by defined sprints, which helped manage the project scope. Regular Daily Scrums were effective for identifying immediate blockers within subgroups.

6.2.2 Challenges and lessons

- **Coordination & Communication:** Coordinating schedules for meetings among a large group proved challenging, especially given varying academic commitments. Ensuring consistent understanding and interpretation of tasks across different subgroups required significant effort and highlighted the need for extremely clear task definitions and regular cross-team communication beyond the Daily Scrum.
- **Consistency:** Maintaining consistency in UI design, coding style, and documentation across different subgroups was an ongoing challenge. This led to extra effort during integration and review phases. Lessons learned include the importance of establishing clear coding standards, design guidelines (like the Style Guide referenced by NFR5), and documentation templates early in the project.
- **Tool Usage:** While Jira and GitHub were used, ensuring everyone utilised them consistently and effectively for task updates and version control required reinforcement.
- **Scope Management:** Balancing the desired feature set (informed by user research) with the time constraints of the sprints required careful prioritisation by the Product Owner and realistic commitments from the development team.

6.2.3 Process Improvements

Based on our experience, we identified several potential process improvements:

- **Enhanced Pre-Planning:** Dedicating more time upfront to establish detailed coding standards, UI style guides, and documentation templates could reduce inconsistencies later.
- **Granular Task Breakdown:** Breaking down backlog items into smaller, more clearly

defined subtasks during Sprint Planning could improve estimation accuracy and make progress tracking easier for individuals.

- **Structured Peer Reviews:** Implementing a formal code review process before merging branches could improve code quality, ensure adherence to standards, and facilitate knowledge sharing.
- **Test Automation:** Introducing automated unit tests and potentially integration tests could improve testing efficiency, provide faster feedback on code changes, and help catch regressions, addressing the limitations noted in the Testing Critique.
- **Communication Tools:** Exploring more structured communication channels or meeting formats might help mitigate scheduling conflicts and ensure information is disseminated effectively across all subgroups.

6.3 Conclusion

The Calorie-Wise project successfully applied Agile principles to develop a PI system prototype tailored for university students. Through iterative development, user research, and team collaboration, we created a functional application addressing core needs like calorie, water, and exercise tracking. While challenges in coordination and consistency were encountered, the Agile process provided the framework to manage these issues and deliver working software. The project highlighted the importance of clear requirements, robust design, thorough testing, and continuous process refinement. Future work should focus on addressing the identified limitations, particularly regarding testing completeness, database scalability, and implementing planned features like enhanced visualisations and streaks, to further improve the system's effectiveness as a student-focused PI tool.

Despite the challenges encountered, the team agrees that the Agile methodology, particularly the iterative nature of Scrum, was beneficial for this project. We would adopt the method again for future projects, incorporating the process improvements identified above, such as enhanced pre-planning and structured peer reviews, to mitigate the coordination and consistency issues faced this time.

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Appendix

Appendix 1: Group Contribution Form (GCF)

Name	Email	Contribution	Comment
Abdollah Aali Hosseini	aah221@bath.ac.uk	10	Took on a leadership role by coordinating the team, organising and leading meetings, managing task distribution, and ensuring consistent progress through regular review of completed work.
Amir Sarwar-Skuse	ahss24@bath.ac.uk	6	Good group member who completed the tasks that were assigned to them.
Eliot Faye	ef693@bath.ac.uk	6	Good group member who completed tasks that were assigned to them.
Kiran Manoj	km2501@bath.ac.uk	6	Good group member who completed the tasks that were assigned to them.
Kourosh Saeidi	ks2805@bath.ac.uk	6	Good group member who completed the tasks that were assigned to them.
Malik Abi Ayad	ma3411@bath.ac.uk	8	Great group member who did the tasks assigned to them, and also went out to accomplish more than expected of them.
Mohammad Omer	mo632@bath.ac.uk	6	Good group member who completed the tasks that were assigned to them.
Nadeem Tabche	nt785@bath.ac.uk	8	Great group member who did the tasks assigned to them, and also went out to accomplish more than expected of them.
Tahmid Ahmed	ta979@bath.ac.uk	10	Took on a leadership role by coordinating the team, organising and leading meetings, managing task distribution, and ensuring consistent progress through regular review of completed work.
Toni Aiyeola	oa748@bath.ac.uk	6	Good group member who completed the tasks that were assigned to them.

Table 1 for Group Contributions and Peer Rated Performance

Appendix 2: Record of meetings

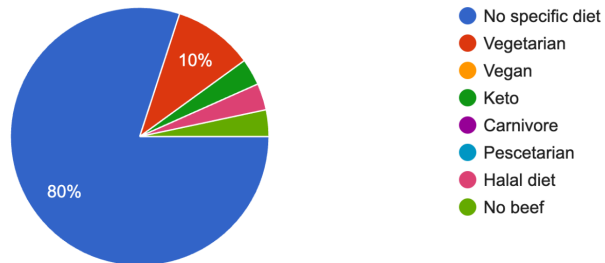
Member Attended	Date Attended	Duration of Meeting	Topics Discussed
Abdollah, Nadeem, Malik, Tahmid, Omer, Toni, Kourosh, Kiran	23/03/2025	2:00pm -> 3:00pm 1 hour	Sprint 1- Division into subgroups Task assignment
Abdollah, Malik, Amir, Kiran, Kourosh, Tahmid, Tahmid	24/03/2025	12:20pm -> 1:20pm 1 hour	Sprint 1 - Task Progress check // Daily SCRUM Meeting
Abdollah, Malik, Nadeem, Kiran, Kourosh, Tahmid	01/04/2025	10:00am -> 11:00am 1 hour	Sprint 1 - Review and retrospectives Sprint 2 - Tasks assigned to subgroups
Abdollah, Malik, Kiran, Eliot, Toni, Tahmid	02/04/2025	12:15pm -> 2:15pm 2 hours	Sprint 2 - Task Progress check // Daily SCRUM Meeting
Abdollah, Malik, Kourosh, Eliot, Kiran, Tahmid, Toni, Omer	05/04/2025	12:00pm -> 1:00pm 1 hour	Sprint 2 -Week 1 Review. Assign next set of tasks
Abdollah, Nadeem, Malik, Tahmid, Omer, Toni, Kourosh, Kiran, Amir, Eliot	06/04/2025	2:00pm -> 3:00pm 1 hour	Sprint 2 - Task Progress check // Daily SCRUM Meeting
Abdollah, Malik, Nadeem, Tahmid, Kiran, Omer	14/04/2025	12:00pm -> 1:00pm 1 hour	Sprint 2 - Week 2 Review. Assign final set of tasks from Product Backlog
Abdollah, Nadeem, Malik, Tahmid, Omer, Toni, Kourosh, Kiran, Amir, Eliot	15/04/2025	10:00am -> 11:00am 1 hour	Sprint 2 - Task Progress check // Daily SCRUM Meeting
Abdollah, Nadeem, Malik, Tahmid, Omer, Toni, Kourosh, Kiran, Amir, Eliot	22/04/2025	4:15pm -> 6:15pm 2 hours	Sprint 2 - Review of Sprint
Abdollah, Nadeem, Malik, Tahmid, Omer, Toni, Kourosh, Kiran, Amir, Eliot	23/04/2025	1:15pm -> 2:15pm 1 hour	Sprint 2 - Sprint Retrospective and what could have been done Report - Assign Tasks
Abdollah, Nadeem, Malik, Tahmid, Omer, Toni, Kourosh, Kiran, Amir, Eliot	26/04/2025	12:00pm -> 2:00 pm 2 hours	Report - Progress check

Table 2 for Meetings

Appendix 3: Research

Do you follow any specific diet?

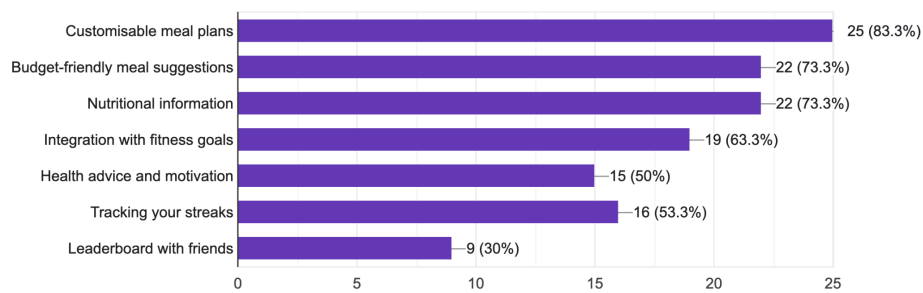
30 responses



Survey Figure 1

What features would you like to see in a meal planning website? (Check all that apply, or add other option)

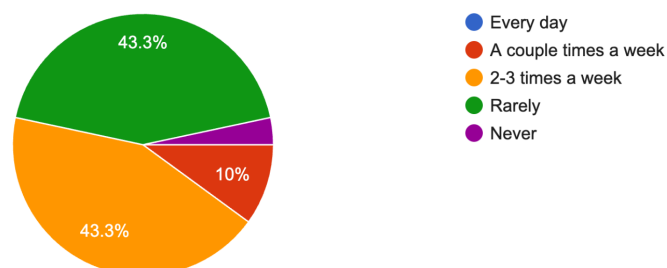
30 responses



Survey Figure 2

How often do you eat out/order takeout?

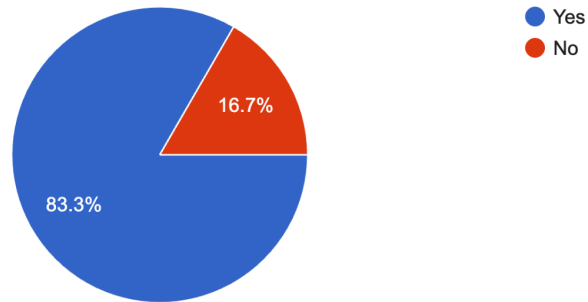
30 responses



Survey Figure 3

Do you have fitness goals?

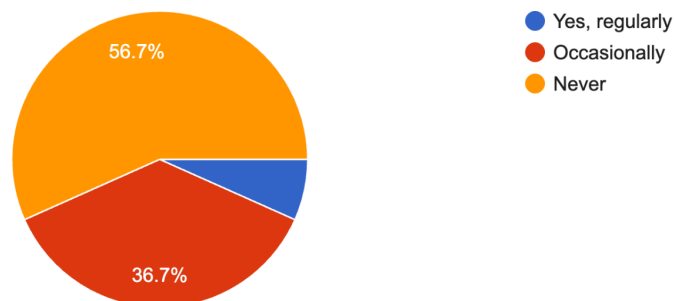
30 responses



Survey Figure 4

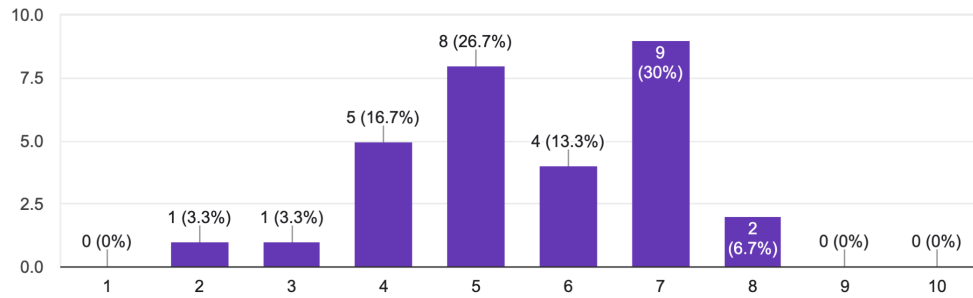
Do you track your calorie intake?

30 responses



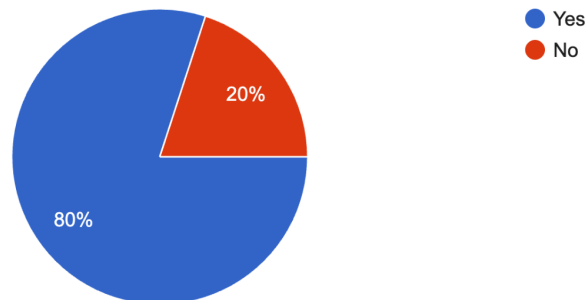
Survey Figure 5

How satisfied are you currently with your diet at University?
30 responses



Survey Figure 6

Do you usually skip any meals?
30 responses



Survey Figure 7

What are the issues you struggle most with in university regarding food?

16 responses

same foods every day, prices
Overpriced
Eating at right times
good options on campus
Getting enough protein
Finding time to cook
Protein intake
Proper meals
Meal planning

Survey Figure 8